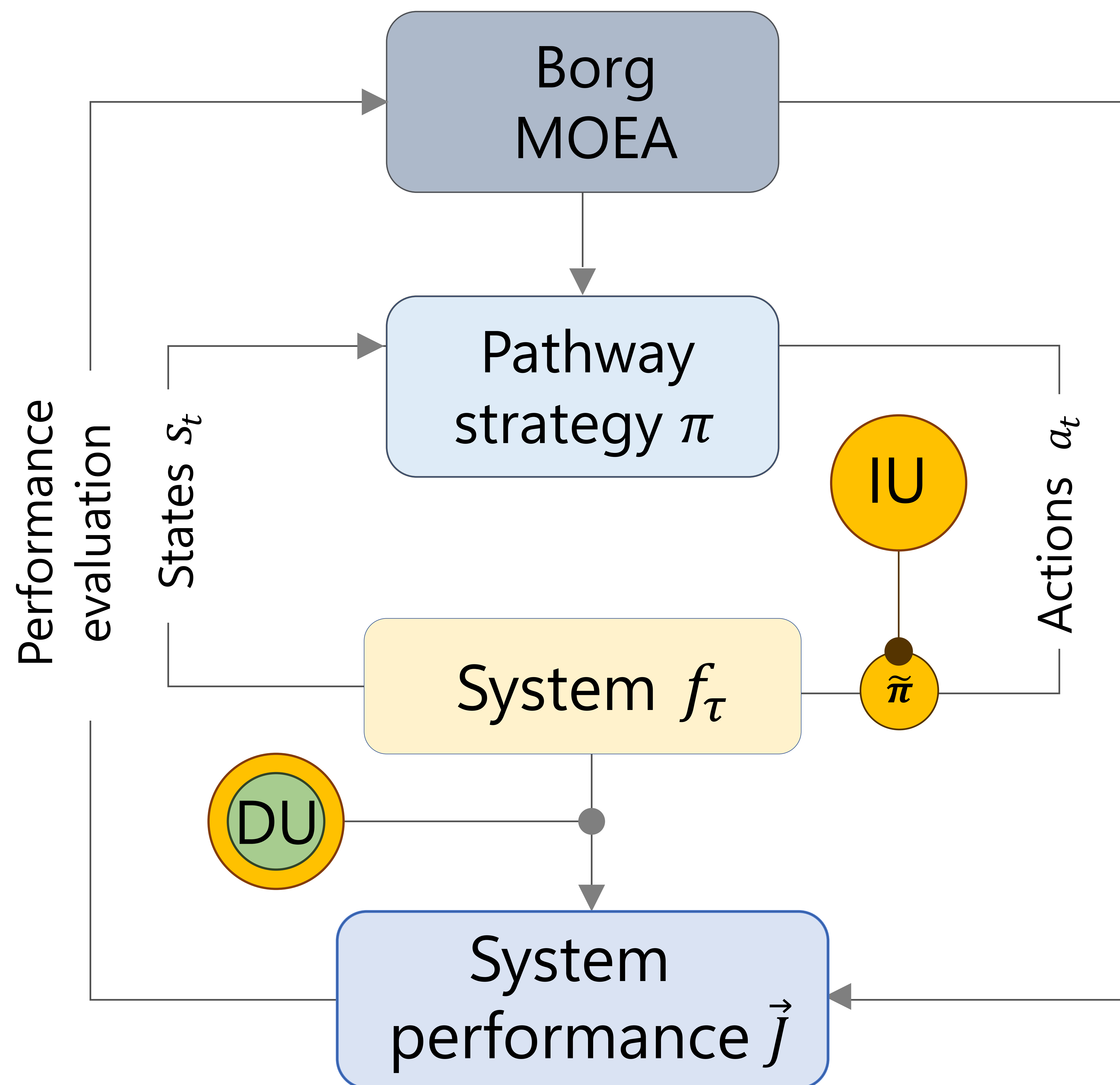
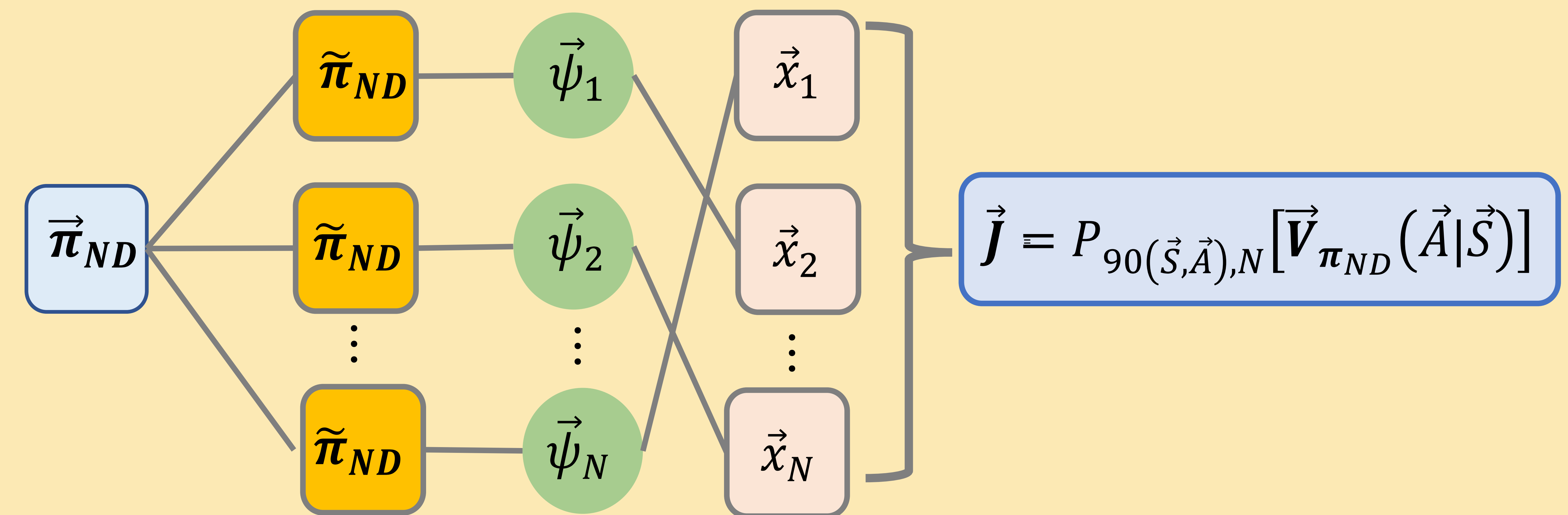


(a) Problem Framing



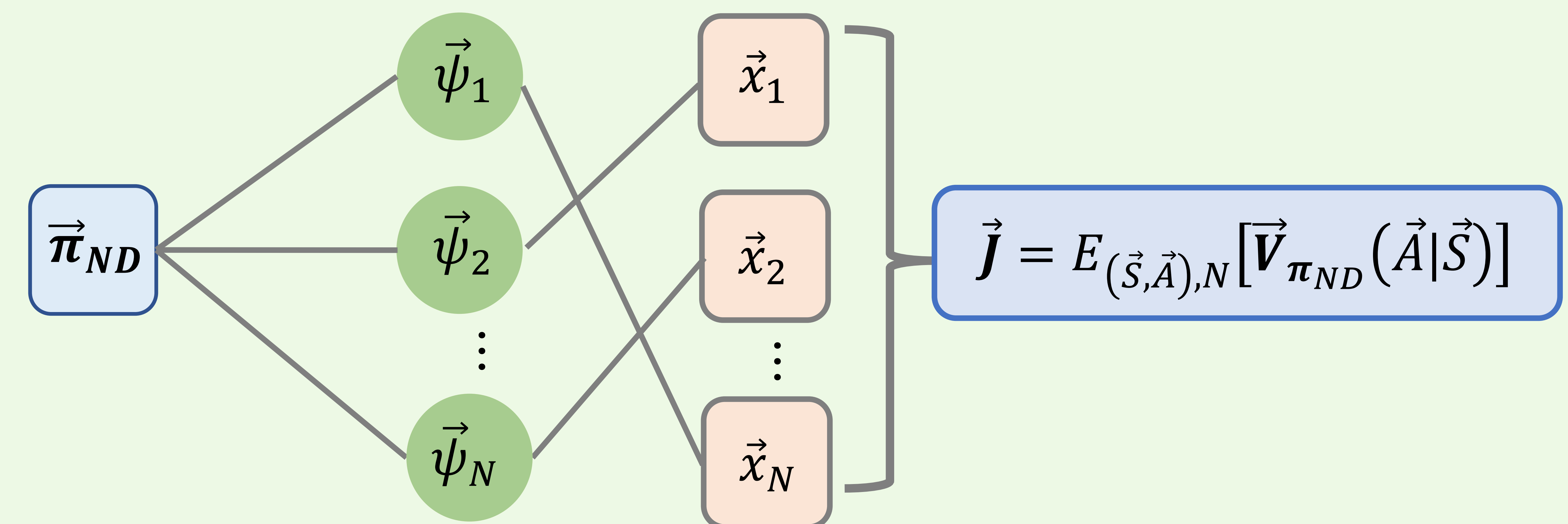
(b) IU-DU Optimization

Worst 10th percentile (WC10) performance objective formulation with implementation uncertainty



(c) Baseline DU Optimization

Expected (EXP) performance objective formulation without implementation uncertainty



Notation

$\vec{\pi}_{ND}$ Original Pareto-approximate strategy

$\tilde{\pi}_{ND}$ Perturbed Pareto-approximate strategy

$\vec{V}_\pi$ Value of a set of states and actions

$\vec{j}$ Performance objective values of a strategy

N Number of uncertain samples

$\vec{\psi}_i \in \vec{\Psi}$ Deeply uncertain state of the world

$\vec{x}_i \in \vec{X}$ Hydroclimatic realization sample

(S, A) Overall set of states and actions

● Baseline DU Optimization

● IU-DU Optimization

DU Deep uncertainty

IU Implementation uncertainty